

Animesh Barua

Software Engineer II · Amazon Web Services

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EXPERIENCE

Software Engineer II · Amazon Web Services

Dec 2024 — Present

Berlin, Germany

- Make AI agents effective on AWS.
 - **MCP servers** — Model Context Protocol servers that let agents work against real AWS infrastructure with least-privilege guardrails, shipped in the open-source [awslabs/mcp](#) project. Primary contributor to the remote [AWS Knowledge MCP Server](#), a fully managed, generally available server giving agents real-time access to AWS documentation, API references, Well-Architected and troubleshooting guidance, regional availability for APIs and CloudFormation resources, and agent skills — with no local setup.
 - **Agent performance** (*ongoing*) — improve how well agents perform on real AWS work: diagnosing misconfigurations, provisioning infrastructure, and operating live cloud environments. Identify and curate the AWS tasks agents fail on, turning those gaps into the datasets that direct evaluation and improvement. Contribute to [aws-bench](#), an open-source benchmark scoring agent and model combinations against disposable, real AWS accounts rather than static fixtures — deploying CDK scenarios, running agents in sandboxed containers, and verifying against live AWS state.

Software Engineer I · Amazon Web Services

Sep 2022 — Nov 2024

Berlin, Germany

- Built [Diagnose with Amazon Q](#) in [Amazon Q Developer](#) — a one-click agent on AWS Management Console errors that explains the root cause in natural language and returns ordered steps to resolve it, replacing the cross-console hunt that multi-service misconfigurations normally require. Featured in the **AWS re:Invent 2023 keynote**.
- Built [resource introspection](#) in Amazon Q Developer using the Strands SDK — an [agentic experience](#) in the AWS Management Console, Slack, and Teams that answers questions about a customer's live AWS environment: breaking a query into executable steps, selecting tools from AWS APIs across **200+ services**, and combining results to explain resource relationships and configuration — within the caller's own IAM permissions.
- Worked on a managed development environment service that helps power AWS products including the [AWS CloudShell](#) IDE.

Software Engineer Intern · Huawei

May 2021 — Jul 2021

Bangalore, India

- Built an ad recommendation system for the ads division on DIEN, a deep-learning CTR model, using TensorFlow and the DeepCTR package.
- Identified and resolved a cold-start problem in the base model, improving AUC from 0.5 — no better than random — to **0.57**.
- Analysed the dataset and devised a data extraction algorithm to obtain the rich user-history data needed to train the model, improving its complexity from **O(n²) to O(n)** and reaching a final AUC of **0.6863**.

EDUCATION

B.Tech, Computer Science and Engineering · IIT Kharagpur

2018 — 2022

Kharagpur, India · CGPA 7.98/10

- **Graph and Geometric Algorithms**, Algorithms-2 term project under Prof. Arobinda Gupta — implemented Ford–Fulkerson for maximum flow in C, extended it to maximise need-based flow across producer–consumer nodes, and adapted Graham's Scan to compute the area enclosed by a territory of circular-shaped cities. [Repository](#).
- Coursework across algorithms, operating systems, computer networks, database management systems, machine learning, and computer architecture.

SKILLS

Languages	Python · TypeScript · C/C++ · Java · Kotlin
Cloud & Infra	AWS · CDK · Docker · CI/CD · OpenTelemetry
AI & Agents	MCP · Agentic AI · Strands SDK · LLMs · Agent evaluation · RAG · LangChain

ACHIEVEMENTS

- Global rank **281** (Round E) and 943 (Round G) in **Google Kick Start 2021**.
- Completed all five levels of Google's Foobar challenge; qualified for Round 2 of Facebook Hacker Cup 2021.

BEYOND WORK

- **Responsibility** — Assistant Technical Coordinator, Technology Students' Gymkhana, IIT Kharagpur: helped overhaul the Gymkhana website and added channels authorising societies to post events. Mentored four first-year students under the Student Welfare Programme.
- **Music** — Qualified the Madhyama examination in Instrumental Violin, Bhatkhande Sangit Vidyapith, Lucknow. Currently learning the guitar.
- **Social & sport** — NSS volunteer: taught underprivileged children and organised cleanliness-awareness and blood-donation camps. Semi-finalist with the Azad Hall football team in the Inter Hall General Championship.